

Problem solving and testing using java

WEEK 1 TASKS

(20/7/2026-25/7/2026)

NAME : VINOTH N

VTUNO : VTU27422

Task 1: Access and print the element at a given index in an array.

Task 2: Search for a given element in a sorted array using Binary Search.

Task 3: Find the maximum element in an array of n integers.

Task 4: Find the K th smallest element in an array.

Task 5: Print all possible pairs of elements from an array.

Task 6: Sum of even or odd digits.

Task 7: N th Fibonacci.

Task 8: Check whether a number is a Palindrome.

Task 9: Sum of the last digits of two given numbers.

TASK : 1

```
package lambda_function; import
```

```
java.util.Scanner;
```

```
@FunctionalInterface    interface
```

```
ArrayElement {    void display(int[]
```

```
arr, int index);
```

```
}
```

```
public class TASK1 {    public static void
```

```
main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    System.out.print("Enter  array  size:  ");
```

```
    int n = sc.nextInt();
```

```
    int arr[] = new int[n];
```

```
    System.out.println("Enter array elements:");
```

```
    for (int i = 0; i < n; i++) {
```

```
arr[i] = sc.nextInt();
```

```
    }
```

```
    System.out.print("Enter  index:  ");
```

```
    int index = sc.nextInt();
```

```

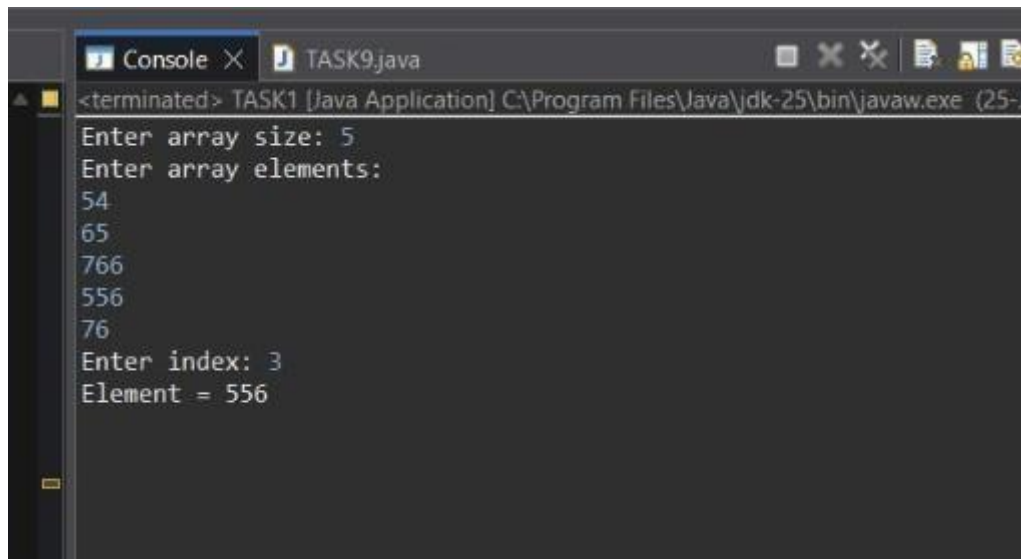
// Lambda Expression
ArrayElement obj = (a, i) -> {
    if (i >= 0 && i < a.length)
        System.out.println("Element = " + a[i]);
    else
        System.out.println("Invalid Index, that index is not there");
};

obj.display(arr, index);

// sc.close();
}
}

```

OUTPUT:



```

Console X TASK9.java
<terminated> TASK1 [Java Application] C:\Program Files\Java\jdk-25\bin\javaw.exe (25-)
Enter array size: 5
Enter array elements:
54
65
766
556
76
Enter index: 3
Element = 556

```

TASK : 2

```
package    lambda_function;

import     java.util.Scanner;

@FunctionalInterface interface

BinarySearch {           void

search(int[] arr, int key);

}

public class TASK2 {     public static void

main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter array size: ");     int

n = sc.nextInt();     int arr[] = new int[n];

    System.out.println("Enter sorted array elements:");

    for (int i = 0; i < n; i++) {

arr[i] = sc.nextInt();

    }

    System.out.print("Enter element to search: ");

int key = sc.nextInt();

    // Lambda Expression

    BinarySearch bs = (a, k) -> {

int low = 0;           int high =

a.length - 1;           boolean

found = false;           while (low

<= high) {           int mid = (low

+ high) / 2;

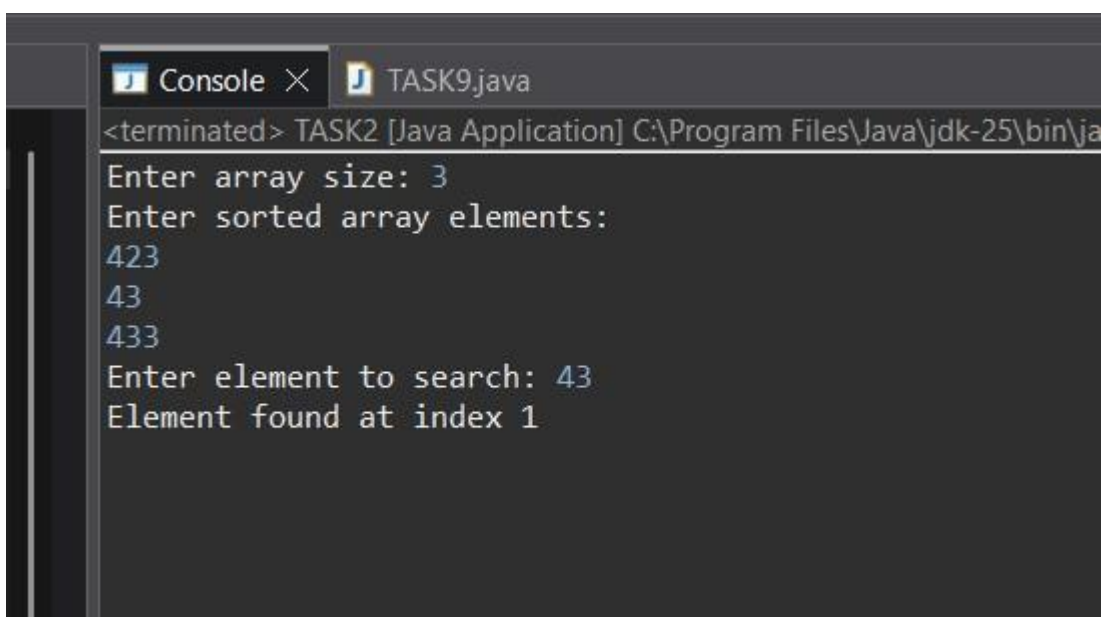
        if (a[mid] == k) {
```

```

        System.out.println("Element found at index " + mid);
found = true;          break;
    } else if (a[mid] < k) {
low = mid + 1;
    } else {
high = mid - 1;
    }
}
}
if (!found) {
    System.out.println("Element not found");
}
};
bs.search(arr,      key);
sc.close();
}
}

```

OUTPUT :



```

Console X TASK9.java
<terminated> TASK2 [Java Application] C:\Program Files\Java\jdk-25\bin\ja
Enter array size: 3
Enter sorted array elements:
423
43
433
Enter element to search: 43
Element found at index 1

```

TASK : 3

```
package lambda_function;

import java.util.Scanner;

@FunctionalInterface

interface Maximum {

    void findMax(int[] arr);

}

public class TASK3 {    public static void

main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter array size: ");    int

n = sc.nextInt();    int arr[] = new int[n];

    System.out.println("Enter array elements:");

    for (int i = 0; i < n; i++) {

arr[i] = sc.nextInt();

    }

    //    Lambda    Expression

    Maximum maxElement = (a) -> {

        int max = a[0];        for (int i

= 1; i < a.length; i++) {            if

(a[i] > max) {                max = a[i];

        }

    }

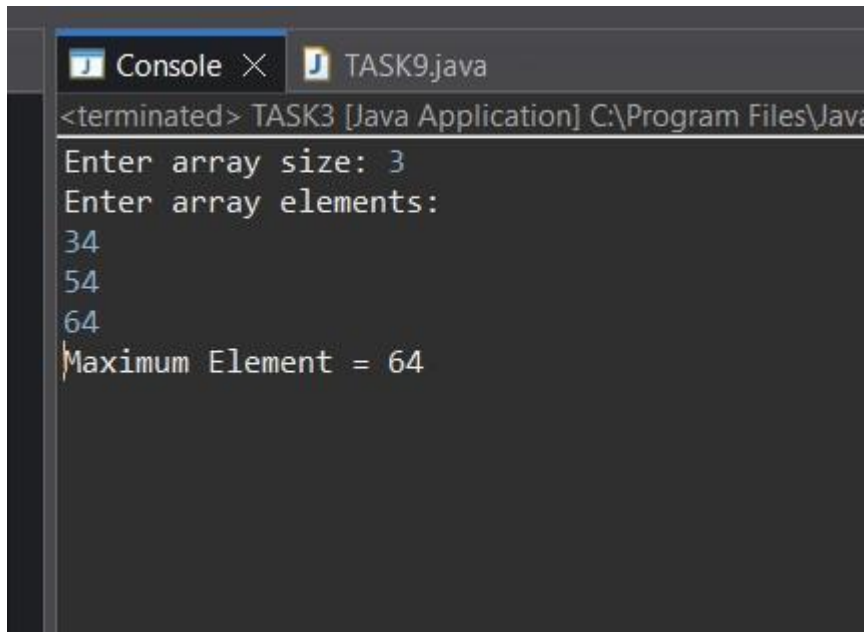
    System.out.println("Maximum Element = " + max);    };

    maxElement.findMax(arr);

    sc.close();
```

```
}  
}
```

OUTPUT:



```
<terminated> TASK3 [Java Application] C:\Program Files\Java  
Enter array size: 3  
Enter array elements:  
34  
54  
64  
Maximum Element = 64
```

TASK : 4 package

lambda_function;

import


```

java.util.Arrays;                import
java.util.Scanner;

@FunctionalInterface              interface
KthSmallestElement {              void
    findKthSmallest(int[] arr, int k);
}

public class TASK4 {    public static void
    main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");
        int n = sc.nextInt();

        int arr[] = new int[n];

        System.out.println("Enter array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.print("Enter K: ");
        int k = sc.nextInt();

        // Lambda Expression
        KthSmallestElement obj = (a, key) -> {    Arrays.sort(a);

            if (key >= 1 && key <= a.length) {

```

```

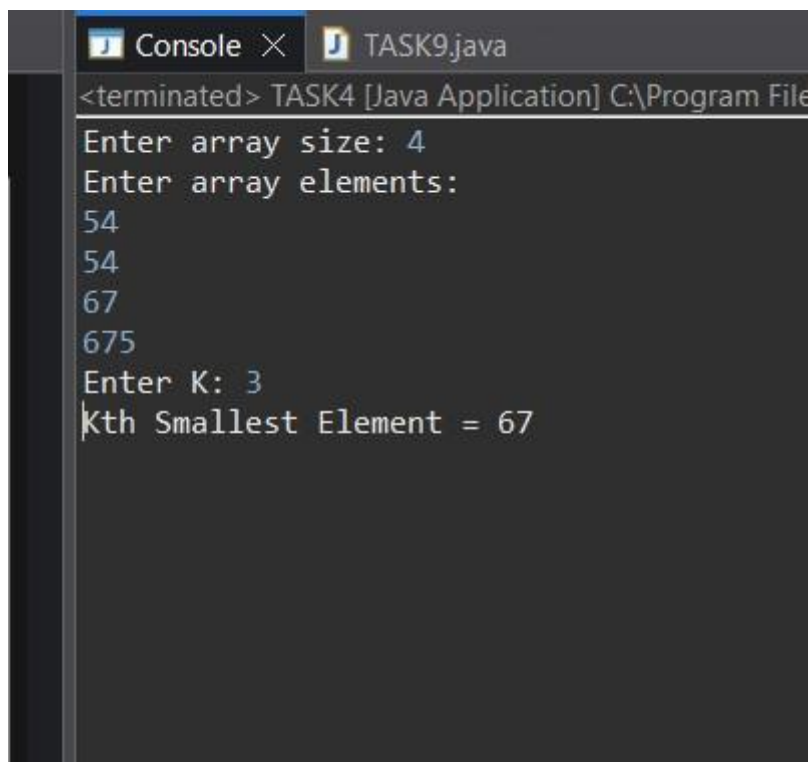
        System.out.println("Kth Smallest Element = " + a[key - 1]);
    } else {
        System.out.println("Invalid K");
    }
};

obj.findKthSmallest(arr, k);

sc.close();
}
}

```

OUTPUT :



```

Console × TASK9.java
<terminated> TASK4 [Java Application] C:\Program File
Enter array size: 4
Enter array elements:
54
54
67
675
Enter K: 3
Kth Smallest Element = 67

```

TASK : 5 package

lambda_function; import

```

java.util.Scanner;

@FunctionalInterface

interface ArrayPair {    void

printPairs(int[] arr);

}

public class TASK5 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter  array  size: ");

int n = sc.nextInt();

        int arr[] = new int[n];

        System.out.println("Enter array elements:");

        for (int i = 0; i < n; i++) {

arr[i] = sc.nextInt();

        }

        //      Lambda      Expression

ArrayPair pair = (a) -> {

        System.out.println("Pairs:");

        for (int i = 0; i < a.length; i++) {

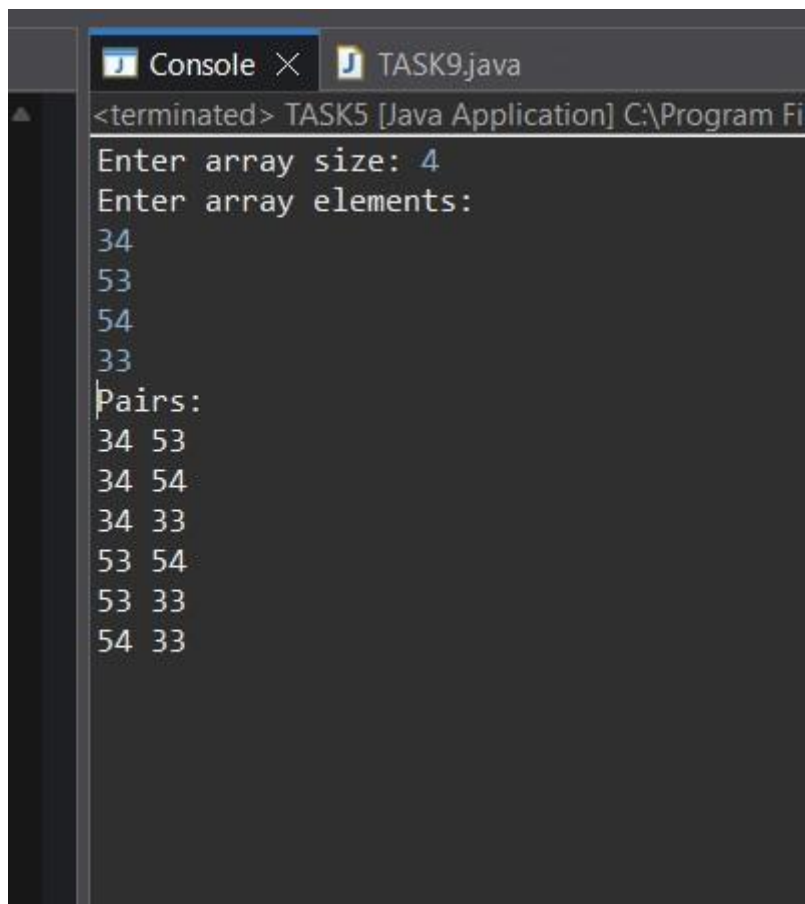
            for (int j = i + 1; j < a.length; j++) {

                System.out.println(a[i] + " " + a[j]);

```

```
        }  
    }  
};  
  
pair.printPairs(arr);  
  
sc.close();  
}  
}
```

OUTPUT :



```
Console x TASK9.java  
<terminated> TASK5 [Java Application] C:\Program Fi  
Enter array size: 4  
Enter array elements:  
34  
53  
54  
33  
Pairs:  
34 53  
34 54  
34 33  
53 54  
53 33  
54 33
```

```
package lambda_function;
```

TASK : 6

```
import java.util.Scanner;
```

```
@FunctionalInterface interface DigitSum {
```

```
void findSum(int num, String choice);
```

```
}
```

```
public class TASK6 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");    int
```

```
num = sc.nextInt();
```

```
        System.out.print("Enter choice (even/odd): ");
```

```
        String choice = sc.next();
```

```
        //      Lambda      Expression
```

```
        DigitSum ds = (number, ch) -> {
```

```
            int sum = 0;    while (number > 0)
```

```
{                int digit = number % 10;
```

```
                if (ch.equalsIgnoreCase("even") && digit % 2 == 0) {
```

```
sum += digit;
```

```
                }
```

```
                if (ch.equalsIgnoreCase("odd") && digit % 2 != 0) {
```

```
sum += digit;
```

```
                }
```

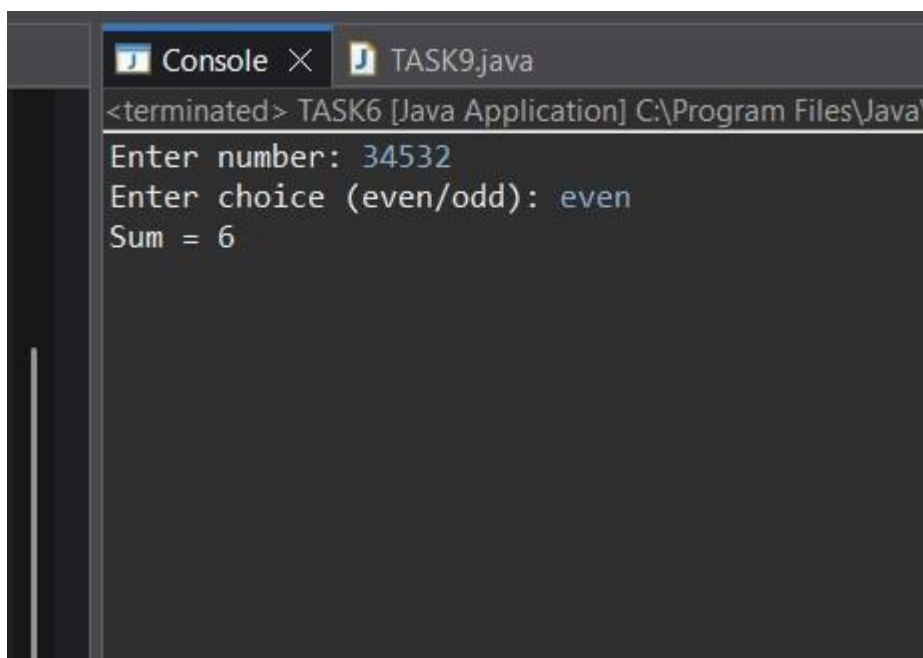
```
        number /= 10;
    }

    System.out.println("Sum = " + sum);
};

ds.findSum(num, choice);

sc.close();
}
}
```

OUTPUT :



```
Console x TASK9.java
<terminated> TASK6 [Java Application] C:\Program Files\Java\
Enter number: 34532
Enter choice (even/odd): even
Sum = 6
```

```
package lambda_function;
```

TASK : 7

```
import java.util.Scanner;
```

```
@FunctionalInterface interface
```

```
FibonacciNumber {    void
```

```
findFibonacci(int n);
```

```
}
```

```
public class TASK7 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter N: ");
```

```
        int n = sc.nextInt();
```

```
        // Lambda Expression
```

```
        FibonacciNumber fib = (num) -> {
```

```
            int a = 0, b = 1;
```

```
            if (num == 1) {
```

```
                System.out.println("Nth Fibonacci = " + a);
```

```
            }
```

```
            else if (num == 2) {
```

```

        System.out.println("Nth Fibonacci = " + b);
    }
    else {
int c = 0;

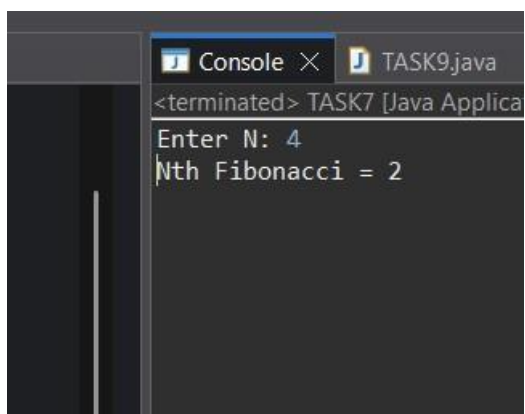
        for (int i = 3; i <= num; i++) {
c = a + b;          a = b;
b = c;
        }

        System.out.println("Nth Fibonacci = " + c);
    }
};

fib.findFibonacci(n);
sc.close();
}
}

```

OUTPUT:



The screenshot shows a Java IDE with a console window titled 'Console' and a file named 'TASK9.java'. The console output shows the program has terminated, followed by the prompt 'Enter N: 4' and the output 'Nth Fibonacci = 2'.

```

<terminated> TASK7 [Java Applica
Enter N: 4
Nth Fibonacci = 2

```

TASK : 8


```
package lambda_function;
```

```
import java.util.Scanner;
```

```
@FunctionalInterface interface
```

```
Palindrome {    void
```

```
checkPalindrome(int num);
```

```
}
```

```
public class TASK8 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
int num = sc.nextInt();
```

```
        // Lambda Expression
```

```
        Palindrome p = (number) -> {
```

```
            int original = number;
```

```
int reverse = 0;
```

```
            while (number > 0) {                int
```

```
digit = number % 10;                reverse
```

```
= reverse * 10 + digit;                number
```

```
 /= 10;
```

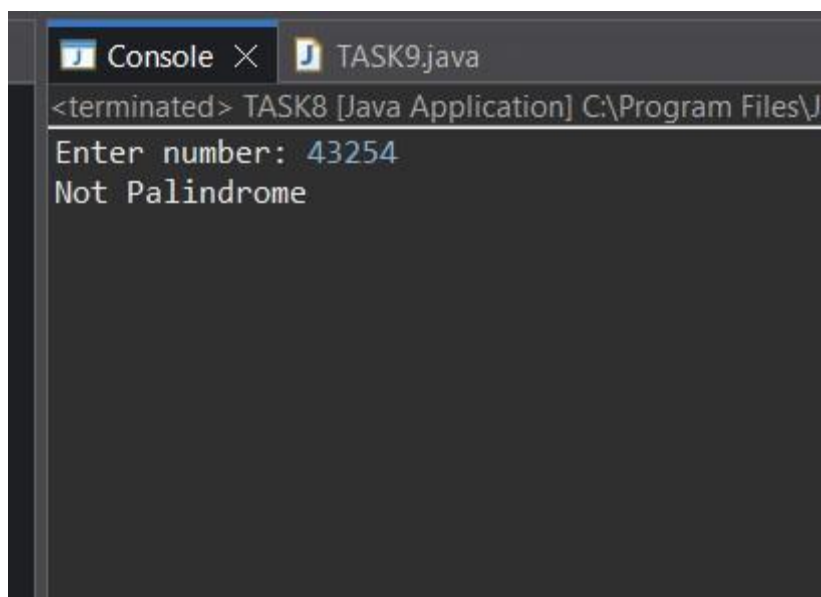
```
        }
```

```
        if (original == reverse)
            System.out.println("Palindrome");
    else
        System.out.println("Not Palindrome");
    };

    p.checkPalindrome(num);

    sc.close();
}
}
```

OUTPUT :



The screenshot shows a Java IDE with two tabs: 'Console' and 'TASK9.java'. The 'Console' tab is active, displaying the following text: '<terminated> TASK8 [Java Application] C:\Program Files\J'. Below this, the user has entered '43254' in response to the prompt 'Enter number:'. The program's output is 'Not Palindrome'.

```
<terminated> TASK8 [Java Application] C:\Program Files\J
Enter number: 43254
Not Palindrome
```

TASK : 9 package

```
lambda_function;
```

```
import java.util.Scanner;
```

```
@FunctionalInterface interface
```

```
LastDigitSum {    void
```

```
sumLastDigits(int a, int b);
```

```
}
```

```
public class TASK9 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter first number: ");
```

```
        int a = sc.nextInt();
```

```
        System.out.print("Enter second number: ");
```

```
        int b = sc.nextInt();
```

```
        // Lambda Expression
```

```
        LastDigitSum sum = (x, y) -> {
```

```
        int last1 = x % 10;

int last2 = y % 10;

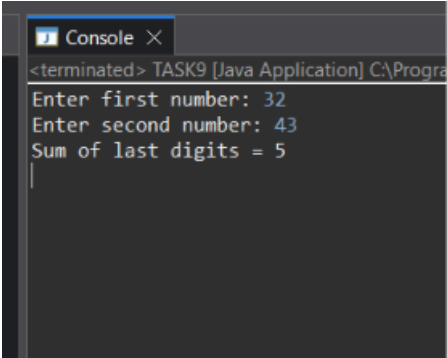
System.out.println("Sum of
last digits = " + (last1 +
last2));

    };

sum.sumLastDigits(a, b);

sc.close();
}
}
```

OUTPUT :



```
Console X
<terminated> TASK9 [Java Application] C:\Progra
Enter first number: 32
Enter second number: 43
Sum of last digits = 5
|
```